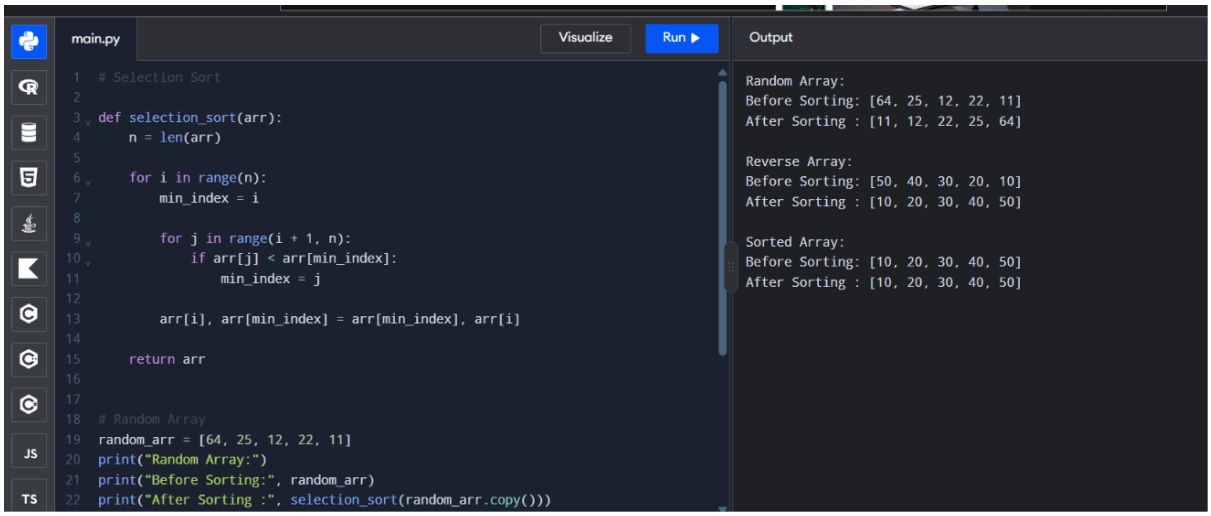
Experiment 7 : Selection Sort on Random, Reverse and Sorted Arrays

Aim

To implement the Selection Sort algorithm and analyze its performance on Random, Reverse, and Sorted arrays.

Algorithm

1. Start with the first element of the array.
2. Assume it is the minimum element.
3. Compare it with the remaining unsorted elements.
4. Find the smallest element in the unsorted portion.
5. Swap the smallest element with the current position.
6. Repeat the process for all positions until the array is sorted.
7. Display the sorted array.



Result

The Selection Sort algorithm was successfully implemented and executed on Random, Reverse, and Sorted arrays. The arrays were sorted in ascending order correctly.

Time Complexity

Case	Time Complexity
Best Case	$O(n^2)$
Average Case	$O(n^2)$
Worst Case	$O(n^2)$